Type 2 Diabetes Risk Prediction with Machine Learning

Project guide based on Xie et al. (Preventing Chronic Disease, 2019) using CDC BRFSS 2014 data

1. Credentials

Item	Detail
Login / account	None needed. The dataset is public.
API key / approval	None needed (unlike MIMIC, which needs credentialing).
Cost	Free

2. The paper

Xie Z, Nikolayeva O, Luo J, Li D. Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques. *Prev Chronic Dis* 2019;16:190109.

Source	Link
DOI	http://dx.doi.org/10.5888/pcd16.190109
Full text (CDC)	https://www.cdc.gov/Pcd/issues/2019/19_0109.htm
Full text (PMC, free)	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6795062/
PDF (CDC Stacks)	https://stacks.cdc.gov/view/cdc/83070

3. Dataset

Item	Detail
Source	CDC Behavioral Risk Factor Surveillance System (BRFSS), 2014
Download page	https://www.cdc.gov/brfss/annual_data/annual_2014.html (link as given in the paper)
Raw size	464,644 respondents, 279 variables
After preprocessing	**138,146 respondents, 20,467 with type 2 diabetes** (about 14.8% positive)
Format	The CDC raw file is usually a SAS transport file (.XPT), not CSV. Pandas can read it (see Setup).

Note: The popular cleaned diabetes CSVs on Kaggle that I know of are built from BRFSS 2015, so they will not match the paper's data exactly. I have not verified any Kaggle version.

4. What the paper did

Step	Detail
Models (8 total)	SVM (linear, polynomial, RBF), decision tree, logistic regression, random forest, Gaussian Naive Bayes, neural network
Class imbalance	SMOTE applied to the training data; testing on the original imbalanced test set
Metrics	Accuracy, sensitivity, specificity, AUC

Step	Detail
Results	AUC ranged from 0.7182 to 0.7949. The neural network had the best AUC; the decision tree had the highest sensitivity (best for screening).
New risk factors	Sleeping 9 or more hours per day; checkup frequency of less than 1 year
Software	R for the models, SAS for the statistics

For a classical-ML-only project, skip the neural network and keep the other 7 models.

5. Setup

```
pip install pandas scikit-learn imbalanced-learn matplotlib seaborn pyreadstat
```

```
import pandas as pd
df = pd.read_sas("LLCP2014.XPT", format="xport")
print(df.shape)  # expect about (464644, 279)
```

The filename LLCP2014.XPT is a guess at what the CDC zip contains. Use whatever file is in the download.

6. Run plan

#	Step
1	Download the 2014 data file from the CDC page and unzip it.
2	Load it and confirm the shape is about 464,644 x 279.
3	Follow the Methods section of the paper for the variable selection and preprocessing that gives the 138,146 respondents.
4	Split into train and test sets, apply SMOTE **only to the training set**, then train the 7 models.
5	Evaluate on the untouched test set with accuracy, sensitivity, specificity, and AUC, and compare with the paper's table.

Details were checked against the paper's CDC and PMC pages. The exact preprocessing steps and feature list are in the paper's Methods section and are not reproduced here.